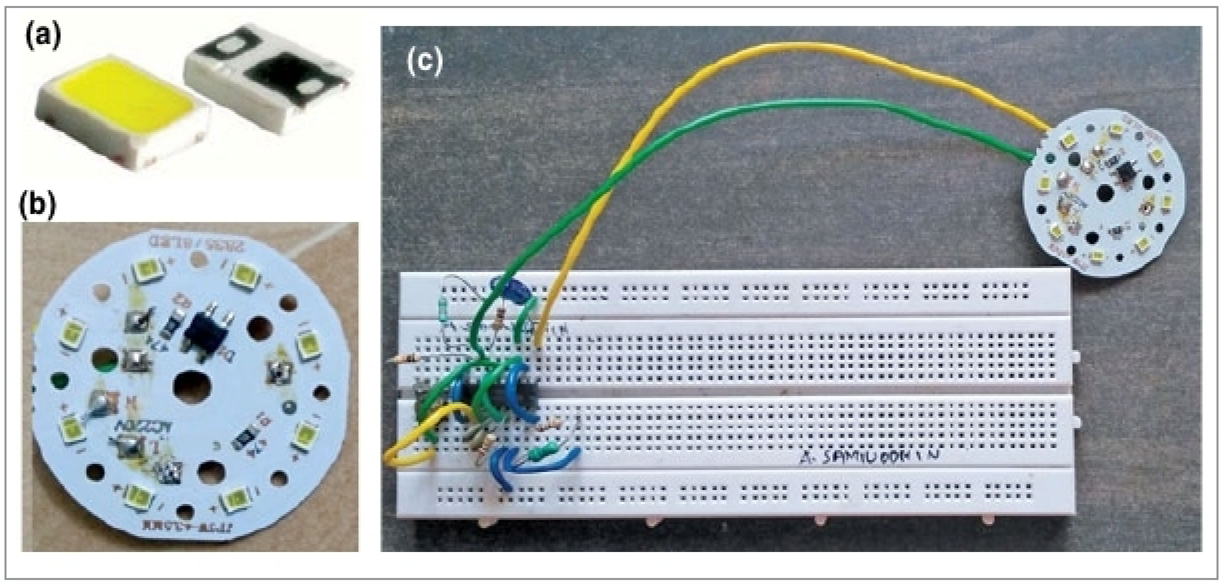

Fig. 4: (a) SMD LED, (b) LED board and (c) author's prototype

IC1, inductor L1 and the onboard circuitry of LED board perform voltage boosting operations. Current regulation is accomplished by monitoring the voltage drop across sense resistor R5, which is placed in series on the LED board. Since maximum voltage across the sense resistor is less than one volt, the on-chip operational amplifier is used to increase the sense voltage for feedback.

Capacitor C1 determines the operating frequency of the circuit. Preset VR1 is used to control the brightness of the LED. Switch S1 is used to switch the LED board on/off.

Construction and testing

An actual-size PCB layout of the night lamp is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a plastic case in such a way that connector CON1 is fixed at the rear side or inside the case. Fix CON2 on the top of the case for connecting the LED board.

A ready-made inductor can be used in place of L1. For better efficiency, an external Schottky diode can be used instead of the internal one. IC 34063 along with an op-amp can be used in the circuit in place of IC1. For peak current greater than 1.5A, an external transistor must be used to avoid device failure.

After assembling and connecting the 12V battery and LED board, your circuit is ready to use.

Use S1 to switch the night bulb on/off. Fig. 4 shows the SMD2835 LED, the LED board with SMD2835 and the author's prototype on the breadboard. EFY

A. Samiuddhin is B.Tech in electrical and electronics engineering. His interests include LED lighting, power electronics, microcontrollers and Arduino programming